

Reasoning Reference: Coding and Decoding

Educational Series

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1 Introduction

Coding and Decoding tests the ability to decipher the rule that transforms a word (message) into a code. Once the rule is identified, it is applied to a new word to find its corresponding code.

2 The Alphabet Reference (Essential)

To solve these quickly, memorize the positions of the English alphabet.

2.1 Forward Order (A=1, Z=26)

Use the keyword **EJOTY** to remember positions: **E=5, J=10, O=15, T=20, Y=25**.

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

2.2 Opposite Pairs (A ↔ Z)

The sum of positions of opposite letters is always **27**. (e.g., A=1, Z=26; $1 + 26 = 27$). Common pairs: **A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N**.

3 Common Coding Patterns

3.1 Letter Shifting

Letters are shifted forward or backward by a constant or increasing number.

- **Example:** If CAT is coded as DBU. (Rule: +1)

3.2 Reverse Order Coding

The word is written in reverse or the letters are replaced by their opposites.

- **Example:** If STEP is PETS. (Rule: Reverse String)
- **Example:** If AZ is ZA. (Rule: Opposite Pairs)

3.3 Numerical/Direct Coding

Each letter is assigned a specific number or symbol.

- **Example:** If APPLE is 1-16-16-12-5. (Rule: Position numbers)

3.4 Chinese Coding (Message Coding)

Groups of words are coded as groups of random words. You identify common words to find their codes.

- sky is blue = pa ma ta
- blue is deep = ma ta za
- (Result: blue/is = ma/ta, sky = pa, deep = za)

4 Solved Examples

Example 1: In a certain code, TEACHER is written as VGCEJGT. How is STUDENT written?

- **Pattern:** $T+2=V$, $E+2=G$, $A+2=C$... (Rule: $+2$)
- **Apply to STUDENT:** $S+2=U$, $T+2=V$, $U+2=W$, $D+2=F$, $E+2=G$, $N+2=P$, $T+2=V$.
- **Answer:** UVWFGPV

Example 2: If $A=1$ and $FAT=27$, then $FAITH = ?$

- **Pattern:** Sum of positions. $F(6) + A(1) + T(20) = 27$.
- **Apply to FAITH:** $F(6) + A(1) + I(9) + T(20) + H(8) = 44$.
- **Answer:** 44